清华大学本科生考试试题专用纸

考试课程：初等概率论　样卷　考试时间：2022 年 * 月 * 日

姓名:_____　学号:_____　院系名称:_____　成绩:_____

请注意：本试卷共 2 页，包括十个大题，满分 100 分.

【可能用到的量或结论：

$\Phi(2.33) = 0.99$; $\Phi(1.64) = 0.95$; $\Phi(1.28) = 0.9$; $\Phi(1) = 0.8413$; $\Phi(0) = 0.5$.

$\Gamma(\alpha, \beta)$ 分布的概率密度函数 $\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} I_{\{x>0\}}$; n 为正整数时，函数 $\Gamma(n) = (n-1)!$.

二元正态分布 $\mathcal{N}(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ 的概率密度函数

$$f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right]\right\}.$$

Slutsky's theorem: 假设 $X_n \xrightarrow{d} X$, $Y_n \xrightarrow{p} c$，则 $X_n + Y_n \xrightarrow{d} X + c$; $X_n Y_n \xrightarrow{d} cX$; $X_n / Y_n \xrightarrow{d} X/c$, $(c \neq 0)$.

Continuous mapping theorem: 设 $\{X, X_n\}$ 是随机元序列，g 连续，则 $X_n \xrightarrow{\text{a.s.}} X \Longrightarrow g(X_n) \xrightarrow{\text{a.s.}} g(X)$; $X_n \xrightarrow{p} X \Longrightarrow g(X_n) \xrightarrow{p} g(X)$; $X_n \xrightarrow{d} X \Longrightarrow g(X_n) \xrightarrow{d} g(X)$.

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一. (10 分) 设 X 是一个随机变量. 对 $1 < x \leq 3$, 其概率密度函数为 $f_X(x) = \frac{x}{4}$; 对其他 x, 其概率密度函数为 $f_X(x) = 0$. 令 $A = \{X \geq 2\}$. 请计算 $f_{X|A}(x), E(X|A)$.

二. (10 分) 判断题：在概率空间 $(\Omega, \mathscr{F}, \mathbb{P})$ 中，假定下述各期望都存在，请判读下列表述是否正确，无需给出理由.

1. (2 分) $P(|X+Y| > 2) \geq \frac{1}{4} E((X+Y)^2)$;
2. (2 分) $E(X^3) \leq \sqrt{E(X^2)E(X^4)}$;
3. (2 分) $E(X^3 Y) = E(X^3 E(Y|X))$.
4. (2 分) 已知 X,Y 不独立，则 X,Y 相关系数不为 0.
5. (2 分) 若 X 与 Y 的分布函数完全一致，即 $F_X(x) = F_Y(x), \forall x \in R$. 则 $X = Y$.

三. (15 分) 设两个学生独立玩某游戏,两人得分都服从一个均匀分布 $Y_i \sim \mathcal{U}(0,1), i = 1, 2$.

(1)(5 分). 令 $V_1 = Y_1 + Y_2$, $V_2 = Y_1 - Y_2$. 请计算相关系数 $\text{corr}(V_1, V_2)$.

(2)(5 分). 求 $\frac{Y_1}{Y_2}$ 的概率密度函数 (PDF).

(3)(5 分). 现假设该游戏不是同时提供给两个人，而是以 $\frac{1}{3}$ 的可能性提供给学生甲，以 $\frac{2}{3}$ 的可能性提供给学生乙，求该游戏的得分 Y 的概率密度函数 (PDF) 和 EY^2.

四. (10 分) 设 X 是一个随机变量，其分布列为

$$p_X(x) = \begin{cases} \frac{x^2}{a}, & \text{若} \quad x = -3, -2, -1, 0, 1, 2, 3, \\ 0, & \text{否则}. \end{cases}$$

(1)(5 分). 求出 a 和 EX.

(2)(5 分). 计算 X 的方差 $\mathrm{var}(X)$.

五. (14 分) 设 $Z_1, ..., Z_n, ...$ 是独立同分布的随机变量序列，且 $Z_1 \sim \mathcal{N}(0, 1)$.

(1)(3 分). 令 $Y = Z_1^2 + ... + Z_n^2$. 求 $E(Y)$.

(2)(3 分). 求 $E\frac{Z_1}{Z_1+Z_2}$.

(3)(4 分). 设 $R^2 = Z_1^2 + Z_2^2$, 求 $\mathrm{cov}(R^2, Z_1)$(需写出具体过程). 进一步，请问 R^2, Z_1 独立么？(若独立，请证明；若不独立，请简要说明)

(4)(4 分). 请利用 Z_1 的矩母函数，求 EZ_1^3, EZ_1^4.

六. (10 分) 设 $X_i \sim B(1, p), i = 1, 2, ...$ 是独立同分布序列，且与 N 独立. 记 $T = X_1 + ... + X_N$.

(1)(5 分). 若 $N \sim \mathcal{P}(\lambda)$ 泊松分布. 求 T 服从什么分布？请给出具体分布名字、参数和推导过程.

(2)(5 分). 若 $N \sim G(s)$ 几何分布. 求 ET.

七. (10 分) 设 $X_1, X_2, ...$ 是独立同分布的随机变量序列. 令 $T_n = X_1 + ... + X_n$.

(1)(5 分) $X_i \sim \mathcal{P}(1)$, 请估计 $P(T_n \leq n)$.

(2)(5 分) $X_i \sim \mathcal{P}(\lambda)$, 记 $M_n(t)$ 为 $\frac{T_n}{n}$ 的矩母函数，请问当 $n \to \infty$ 时，$M_n(t)$ 的极限是什么？

八. (8 分)

设全世界有 n 个家庭，每个家庭有 k 个小孩的概率都是 p_k. 设 p_k 满足 $\sum_{k=0}^{c} p_k = 1$. 如果各个家庭的小孩数是相互独立的，计算一个小孩来自有 k 个小孩的家庭的概率.

九. (5 分) 设一个盒子中有 M 个红球，$N - M$ 个黑球，每个球等可能被抽到. 现从中无放回地任取 n 个，记其中红球个数为 X. 求 $\mathrm{var}(X)$.

十. (8 分)

某个银行选择一天举办发行纪念币活动，并安排了两个员工分别前往到两个不同的网点进行此工作. 设各网点顾客无缝衔接排队购买，每个顾客接受服务的时间都是相互独立，且服从指数分布 $\mathcal{E}(\lambda)$(单位：分钟). 设两人工作状态相互独立. 考虑到通勤时长等因素，公司规定员工 A 的工作时长为 t 分钟，员工 B 的工作时长为 s 分钟. 每位员工在工作时间结束时，若正在服务顾客，则交接给后续接班的其他员工. 记 X 为员工 A 完成的服务人次，Y 为员工 B 完成的服务人次.

(1)(3 分) 请问 X 服从什么分布？请给出具体求解过程.

(2)(3 分) 若已知 A 和 B 共完成服务 100 人，求平均 A 完成服务多少人？请给出具体求解过程.

清华大学本科生考试试题专用纸

Course Title: Probability Theory. Time and Date(UTC+8): 8-10am Dec. 31, 2020

Name:_____ Student ID:_____ Version: Paper A

Note: This exam is closed book and closed notes. There are 10 problems on 2 pages, worth 100 points in total.

【Some information for your reference:

$\Phi(2.33) = 0.99$; $\Phi(1.64) = 0.95$; $\Phi(1.28) = 0.9$; $\Phi(1) = 0.8413$; $\Phi(0) = 0.5$.

PDF of $\Gamma(\alpha, \beta)$ is $\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} I_{\{x>0\}}$; $\Gamma(n) = (n-1)!, n = 1, 2, ...$

PDF of $\mathcal{N}(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ is

$$f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right]\right\}.$$

Slutsky's theorem: Suppose $X_n \xrightarrow{d} X$, $Y_n \xrightarrow{p} c$, then $X_n + Y_n \xrightarrow{d} X + c$; $X_n Y_n \xrightarrow{d} cX$; $X_n / Y_n \xrightarrow{d} X/c$, $(c \neq 0)$.

Continuous mapping theorem: Suppose $\{X, X_n, n = 1, 2, ...\}$ are random variables, g is a continuous function, then $X_n \xrightarrow{\text{a.s.}} X \Longrightarrow g(X_n) \xrightarrow{\text{a.s.}} g(X)$; $X_n \xrightarrow{p} X \Longrightarrow g(X_n) \xrightarrow{p} g(X)$; $X_n \xrightarrow{d} X \Longrightarrow g(X_n) \xrightarrow{d} g(X)$.

Distribution	$B(n,p)$	$\mathcal{P}(\lambda)$	$G(p)$	$\mathcal{U}(0,1)$	$\mathcal{E}(\lambda)$	$\mathcal{N}(\mu, \sigma^2)$
Moment Generating Function	$(1-p+pe^s)^n$	$e^{\lambda(e^s-1)}$	$\frac{pe^s}{1-(1-p)e^s}$	$\frac{e^s-1}{s}$	$\frac{\lambda}{\lambda-s}$	$e^{\mu s+(\sigma^2 s^2/2)}$

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1. (10 points) Let X be a random variable with PDF $f_X(x) = \frac{x}{4}$ for $1 < x \leq 3$ and $f_X(x) = 0$ for other x. Let $A = \{X \geq 2\}$. Find $f_{X|A}(x)$ and $E(X|A)$.

2. (10 points) True or False Questions: In the probability space $(\Omega, \mathscr{F}, \mathbb{P})$, assuming the following expectations exist,

1. (2 points) $P(|X+Y| > 2) \geq \frac{1}{4}E((X+Y)^2)$;
2. (2 points) $E(X^3) \leq \sqrt{E(X^2)E(X^4)}$;
3. (2 points) $E(X^3Y) = E(X^3E(Y|X))$.
4. (2 points) If X and Y are not independent, then $\text{corr}(X, Y) \neq 0$.
5. (2 points) If for random variables X and Y, we have $F_X(x) = F_Y(x), \forall x \in R$, then $X = Y$.

3. (15 points) Suppose there is a game. If Alice play it, the score would be $Y_1 \sim \mathcal{U}(0,1)$. If Bob play it, the score would be $Y_2 \sim \mathcal{U}(0,1)$. Y_1 and Y_2 are independent.

(1)(5 points). Let $V_1 = Y_1 + Y_2$, $V_2 = Y_1 - Y_2$. Find the correlation coefficient $\text{corr}(V_1, V_2)$.

(2)(5 points). Find the probability density function (PDF) of $\frac{Y_1}{Y_2}$.

(3)(5 points). Now the game can only be provided to one person. It is provided to Alice and Bob with probability $\frac{1}{3}$ and $\frac{2}{3}$ separately. Find the PDF of the game score Y and EY^2.

4. (10 points) Let X be a random variable with PMF

$$p_X(x) = \begin{cases} \frac{x^2}{a}, & \text{if} \quad x = -3, -2, -1, 0, 1, 2, 3, \\ 0, & \text{otherwise.} \end{cases}$$

(1)(5 points). Find a and EX.

(2)(5 points). Find the variance of X, i.e. $\text{var}(X)$.

5. (14 points) Let $Z_1, ..., Z_n, ...$ be *i.i.d.* random variables, and $Z_1 \sim \mathcal{N}(0, 1)$.

(1)(3 points). Let $Y = Z_1^2 + ... + Z_n^2$. Find $E(Y)$.

(2)(3 points). Find $E\frac{Z_1}{Z_1+Z_2}$.

(3)(4 points). Let $R^2 = Z_1^2 + Z_2^2$. Find $\text{cov}(R^2, Z_1)$. Further, are R^2 and Z_1 independent? (Please provide your reason or proof)

(4)(4 points). Find EZ_1^3 and EZ_1^4 using the MGF of Z_1.

6. (10 points) Let $X_i \sim B(1, p), i = 1, 2, ...$ be a sequence of independent identically distributed random variables, and independent with N. Let $T = X_1 + ... + X_N$.

(1)(5 points). Let $N \sim \mathcal{P}(\lambda)$. Find the distribution of T.

(2)(5 points). Let $N \sim G(s)$. Find ET.

7. (10 points) Let $X_1, X_2, ...$ be a sequence of independent identically distributed (*i.i.d.*) random variables. Let $T_n = X_1 + ... + X_n$.

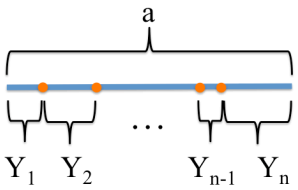
(1)(5 points) $X_i \sim \mathcal{P}(1)$. Find the approximation of $P(T_n \leq n)$.

(2)(5 points) $X_i \sim \mathcal{P}(\lambda)$. Let $M_n(t)$ be the MGF of $\frac{T_n}{n}$. Find $\lim\limits_{n\to\infty} M_n(t)$.

8. (8 points) Suppose there are 2 flights between Nanjing and Beijing. 600 passengers choose either flight equal likely and independently. How many seats should the first flight set up, so that the probability of overbooking on this flight is 0.01?

9. (5 points) Suppose there are M red balls and $N-M$ black balls in a box, with equal probability being picked. Now we pick n balls randomly without replacement, and denote the number of red balls as X. Find $\text{var}(X)$.

10. (8 points) Suppose there is a blue stick with length a. We choose $n-1$ points independently and uniformly on it, and cut the stick at these points. So we get n small sticks, with length $Y_1, ..., Y_n$ as shown in the figure.



(1)(2 points) Find the distribution of Y_1.

(2)(2 points) For $n = 2$, find the expectation of $\max\limits_{1\leq k\leq n} Y_k$.

(3)(4 points) For $n = 3$, find the expectation of $\max\limits_{1\leq k\leq n} Y_k$.